Clinical Decision Support System for Disease Risk Assessment

Progress Report - 4

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1. Work Date / Hours Log

Date	Hours	Description of Work Done
Mar 11, 2026	3.0	Modified Streamlit UI to transition from CBC-based display to disease-focused outputs (diabetes and heart disease).
Mar 13, 2026	3.5	Integrated diabetes ML model with UI and connected extracted indicators (glucose, BMI, blood pressure, age).
Mar 14, 2026	3.5	Updated UI components to display dual outputs: ML-based diabetes risk and rule-based heart disease risk.
Mar 16, 2026	4.0	Collaborated with Akif to test end-to-end system and ensured correct mapping of extracted data to UI outputs.

2. Summary Description of Work Done During This Reporting Period

This reporting period saw the Streamlit application greatly revised by switching to a disease-specific interface (both diabetes and heart disease) instead of a CBC-centered display.

The UI was restructured to show the results of both methods: a machine learning model to predict the risk of diabetes and a rule-based system to predict the risk of heart diseases. The indicators that were structured like the glucose, BMI, blood pressure, and age were used in the diabetes prediction pipeline, and the lipid profile indicators were redirected to the heart disease module.

The interface was also improved to show risk level, confidence score, risk score, and contributing indicators in an understandable format of both conditions. This provided consistency in the processing of the back end and presentation on the front end.

The system was tested together with Akif using synthetic medical reports to confirm end-to-end functionality. The problems concerning the lack of consistency in outputs and data mapping were addressed while debugging and refinement. The app has now been updated to the disease-oriented architecture and can analyse risks.

3. Repo Check-In of Implementation Completed

Application Updates

- `sampleapp.py` (updated Streamlit UI for diabetes and heart disease risk display)

Model Integration

- `diabetes_model_inference.py` (ML model integration into UI)

Collaboration & Testing

- Integration with `disease_pipeline.py` outputs
- End-to-end validation of UI with extracted indicators and risk outputs

4. AI Use Section

AI Tool	Version	Purpose of Use	Value Added by User
ChatGPT	GPT-5 (Premium)	Suggestions for improving interpretability of results	Refined display of contributing indicators and risk explanations
Gamma AI	Free Version	Generating medical reports for diabetes and heart disease	Provided accurate prompt for generation to maintain a clinical level lab report

5. APPENDIX

5.1 Appendix A – AI Prompts

1. How can I present ML predictions in a more interpretable way?

   Create a realistic hospital laboratory report as a single-page document

   The document should look like a medical diagnostic report used in hospitals.

   Include:

   Patient Information Section Patient Name Age Gender Date of Test Laboratory Name

   Create a laboratory test results table with columns:

   Test Name | Result | Unit | Reference Range

   Include these tests:

   Glucose | 132 | mg/dL | 70–99 HbA1c | 6.7 | % | <5.7 Total Cholesterol | 235 | mg/dL | <200 LDL | 160 | mg/dL | <100 HDL | 38 | mg/dL | >40 Triglycerides | 210 | mg/dL | <150 Blood Pressure | 145/92 | mmHg | <120/80

   The final output should resemble a professional hospital lab report layout with clean formatting suitable for printing.

2.

5.2 Appendix B – UI Output Features

The application displays:

- Diabetes Risk (ML Model)
- Heart Disease Risk (Rule-Based)
- Risk Level
- Confidence Score
- Contributing Indicators